

Kubernetes and Cloud Native Associate - KCNA

Container Orchestration Networking

Cluster Networking

Welcome to this comprehensive guide on configuring networking for both master and worker nodes within a Kubernetes cluster. In this article, we detail the necessary network settings and port configurations required for optimal cluster communication and performance.

Node Configuration

In a Kubernetes cluster, each node must have at least one network interface that is configured with:

- A unique IP address.
- A unique hostname.
- A distinct MAC address.

It is particularly important to ensure these settings are correctly applied if virtual machines (VMs) have been cloned from existing ones.

Essential Port Configurations

Proper communication between Kubernetes components relies on opening specific network ports. Below is a list of the critical ports that must be accessible:

- **Port 6443:**

The Kubernetes API server, running on the master node, listens on this port. It must be open for:

- Worker nodes.
- The `kubectl` command-line tool.
- External users.
- Other control plane components.

- **Port 10250:**

The kubelet service runs on both the master and worker nodes and listens on this port for managing node-level operations.

- **Port 10259:**

The kube-scheduler service uses this port to operate smoothly.

- **Port 10257:**

The kube-controller-manager service requires this open port for its functions.

- **Ports 30000 to 32767:**

These ports are used by worker nodes to expose services to external users.

- **Port 2379:**

The etcd server, which handles key-value storage for the cluster, listens on this port.

If you are running multiple master nodes, ensure that **port 2380** is also open to facilitate communication between etcd clients.



Note

For a full list of required ports, please refer to the [official Kubernetes documentation](#).

Network Security Considerations

When setting up your Kubernetes cluster, make sure that the following elements are included in your network security configuration:

- **Firewall Settings:** Ensure that appropriate ports are allowed.
- **iptables Rules:** Confirm that the necessary rules enable smooth communication between components.
- **Cloud Network Security Groups:** For clusters hosted on cloud platforms such as GCP, Azure, or AWS, verify that relevant security groups permit the required ports.

If any component exhibits communication issues, a primary troubleshooting step should be to review and adjust your port configuration settings.



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